Conceptual Frontier Deepening

Expert inference validation report

Focus question
Which conceptual hypothesis survived independent challenge and why?

Run Id
fixture.conceptual_frontier_deepening.base

Target
mathematical

Trace Id
trace.0ca65077522ff37a

Workflow Id
conceptual_frontier_deepening

Workflow Name
Conceptual Frontier Deepening

Workflow Version
1.0.0

Canonical packet SHA-256
bcdde060b940e2d43a801de0a437584b30a625cb3a118547d7c24402e4d38e77

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1 Expert reading contract

Public rationale boundary

Inspectable public rationale and evidence only. The inventory must not contain private chain-of-thought or derive scientific claims from layout, chronology, control authorization, or artifact integrity.

This document records inspectable premises, public warrants, evidence, conclusions, uncertainty, scope, authority, failures, and obligations. It does not expose private chain-of-thought. Execution authorization, artifact integrity, scientific epistemic status, and workflow disposition are independent. Chronological precedence does not mean logical or evidential support.

1.1 Status legend

Artifact integrity: whether referenced bytes and receipts validate **Scientific epistemic status:** the bounded support assigned to a claim **Workflow disposition:** what the workflow did with the item

2 Validation dashboard

Control coverage

1.0

Semantic coverage

1.0

Evidence resolution

1.0

Visible trace gaps

0

Required expert contrasts

["frozen context versus proposed extension", "candidate hypothesis versus blind adjudication", "surviving claim versus rejected alternative"]

3 Declared route and topology

Route spine

Frozen context to formal-ready claim

Topology mode

dag

Step	Skill	Mode / observed state	Consumes	Produces
plan_deepening	conceptual-frontier-deepening	required / observed	["FrozenDiscoveryContext"]	["ConceptualDeepeningPlan"]
assimilate_literature	frontier-literature-assimilator	required / observed	["FrozenDiscoveryContext"]	["LiteratureFrontierMap"]
forge_invention	literature-conditioned-invention-forger	required / observed	["LiteratureFrontierMap"]	["InventionCandidates"]
portfolio	parallel-hypothesis-portfolio	required / observed	["InventionCandidates"]	["HypothesisPortfolio"]
reflect_discovery	reflective-conceptual-discovery	required / observed	["HypothesisPortfolio"]	["ConceptualReflection"]

Step	Skill	Mode / observed state	Consumes	Produces
synthesize	reflective-theory-synthesizer	required / observed	["ConceptualReflection"]	["TheoryCandidates"]
blind_adjudicate	blind-conceptual-preference-adjudicator	required / observed	["TheoryCandidates"]	["BlindAdjudication"]
compile_atlas_defects	theory-atlas-defect-compiler	conditional / observed	["TheoryCandidates"]	["TheoryAtlasDefects"]
derive_central_routes	central-formal-conceptual-routes	required / observed	["BlindAdjudication"]	["CentralConceptualRoutes"]
check_bridge_equivalence	bridge-object-equivalence-checker	conditional / observed	["CentralConceptualRoutes"]	["BridgeEquivalenceDecision"]
produce_formal_claim	formal-claim-producer	required / observed	["CentralConceptualRoutes", "BlindAdjudication"]	["FormalClaimProducerOutputV2"]

4 Profile topology and workflow-specific contrasts

Projection lens

conceptual_frontier_deepening

Topology mode

dag

Required expert contrasts

["frozen context versus proposed extension", "candidate hypothesis versus blind adjudication", "surviving claim versus rejected alternative"]

Declared profile topology

No specialized topology applies

Observed specialist roles

No specialist fan-out applies

A specialist fork or join is a workflow relation, not proof that specialist conclusions are mutually consistent. Finite, endpoint, asymptotic, and continuum boundaries remain separate unless an explicit validated inference connects them.

5 Expert inference dossiers

Each dossier is bound to its semantic event ID. Control-only events appear in the lifecycle appendix and are not scientific claims.

5.1 Inference 1: conceptual_frontier_deepening.base.plan_deepening.inference

---

Trace disposition

Complete public dossier

Step / skill

plan_deepening / conceptual-frontier-deepening

Question

At plan_deepening, how does FrozenDiscoveryContext bear on ConceptualDeepeningPlan? Which representation resolves the repeated bridge-object blocker under blind comparison?

Typed premises

[{"statement": "FrozenDiscoveryContext: The frozen context contains two non-dominated carrier constructi ons.", "type": "artifact"}]

Transformation / operator

artifact_grounded_transformation

Bounded public warrant

At plan_deepening, artifact_grounded_transformation checks FrozenDiscoveryContext before emitting ConceptualDeepeningPlan. Authority boundary: May propose and rank theories under frozen context; experiment s constrain but never select a conceptual theory, and no proof is certified.

Conclusion

plan_deepening produces ConceptualDeepeningPlan: Retain the quotient carrier as a formal-claim candidate.

Uncertainty

bounded by the declared artifact and open obligations

Assumptions

None recorded

Scope before / after / model / regime

{"after": "formal-claim candidate", "before": "conceptual candidate", "model": "competing carriers", "regime": "frozen theory context"}

Causal parents

["conceptual_frontier_deepening.base.plan_deepening.control"]

Artifact integrity status

integrity_valid

Scientific epistemic status

evidence_backed

Workflow disposition

accepted

Evidence links

evidence.d20af4ce1f3d92

Alternative / falsifier / failure links

alternative.08f9b00fdf4262, falsifier.4e30d6237d375c

Obligation links

None recorded

Artifact / receipt links

None recorded

Authority

{"absent": ["unscoped-epistemic-upgrade"], "exercised": ["plan_deepening-assessment"], "prohibitions_checked": ["no hidden scope promotion"]}

Handoff

None recorded

5.2 Inference 2: conceptual_frontier_deepening.base.assimilate_literature.inference

---

Trace disposition

Complete public dossier

Step / skill

assimilate_literature / frontier-literature-assimilator

Question

At assimilate_literature, how does FrozenDiscoveryContext bear on LiteratureFrontierMap? Which representa tion resolves the repeated bridge-object blocker under blind comparison?

Typed premises

[{"statement": "FrozenDiscoveryContext: The frozen context contains two non-dominated carrier constructi ons.", "type": "artifact"}]

Transformation / operator

assimilate_prior_evidence

Bounded public warrant

At assimilate_literature, assimilate_prior_evidence checks FrozenDiscoveryContext before emitting Literature FrontierMap. Authority boundary: May propose and rank theories under frozen context; experiments constrain but never select a conceptual theory, and no proof is certified.

Conclusion

assimilate_literature produces LiteratureFrontierMap: Retain the quotient carrier as a formal-claim candidate.

Uncertainty

bounded by the declared artifact and open obligations

Assumptions

None recorded

Scope before / after / model / regime

{"after": "formal-claim candidate", "before": "conceptual candidate", "model": "competing carriers", "regime": "frozen theory context"}

Causal parents

["conceptual_frontier_deepening.base.assimilate_literature.control"]

Artifact integrity status

integrity_valid

Scientific epistemic status

evidence_backed

Workflow disposition

accepted

Evidence links

evidence.0697c302322c1f

Alternative / falsifier / failure links

alternative.09c22f7abfdc2d, falsifier.f46d99fc154bf7

Obligation links

None recorded

Artifact / receipt links

None recorded

Authority

{"absent": ["unscoped-epistemic-upgrade"], "exercised": ["assimilate_literature-assessment"], "prohibitions_ checked": ["no hidden scope promotion"]}

Handoff

None recorded

5.3 Inference 3: conceptual_frontier_deepening.base.forge_invention.inference

Trace disposition

Complete public dossier

Step / skill

forge_invention / literature-conditioned-invention-forger

Question

At forge_invention, how does LiteratureFrontierMap bear on InventionCandidates? Which representation resolves the repeated bridge-object blocker under blind comparison?

Typed premises

[{"statement": "LiteratureFrontierMap: The frozen context contains two non-dominated carrier constructions. ", "type": "artifact"}]

Transformation / operator

assimilate_prior_evidence

Bounded public warrant

At forge_invention, assimilate_prior_evidence checks LiteratureFrontierMap before emitting InventionCandidates. Authority boundary: May propose and rank theories under frozen context; experiments constrain but never select a conceptual theory, and no proof is certified.

Conclusion

forge_invention produces InventionCandidates: Retain the quotient carrier as a formal-claim candidate.

Uncertainty

bounded by the declared artifact and open obligations

Assumptions

None recorded

Scope before / after / model / regime

{"after": "formal-claim candidate", "before": "conceptual candidate", "model": "competing carriers", "regime": "frozen theory context"}

Causal parents

["conceptual_frontier_deepening.base.forge_invention.control"]

Artifact integrity status

integrity_valid

Scientific epistemic status

evidence_backed

Workflow disposition

accepted

Evidence links

evidence.c80c8c4a3127a7

Alternative / falsifier / failure links

alternative.a83038138ba717, falsifier.a2eecd714d4e86

Obligation links

None recorded

Artifact / receipt links

None recorded

Authority

{"absent": ["unscoped-epistemic-upgrade"], "exercised": ["forge_invention-assessment"], "prohibitions_checked": ["no hidden scope promotion"]}

Handoff

None recorded

5.4 Inference 4: conceptual_frontier_deepening.base.portfolio.inference

---

Trace disposition

Complete public dossier

Step / skill

portfolio / parallel-hypothesis-portfolio

Question

At portfolio, how does InventionCandidates bear on HypothesisPortfolio? Which representation resolves the repeated bridge-object blocker under blind comparison?

Typed premises

[{"statement": "InventionCandidates: The frozen context contains two non-dominated carrier constructions.", "type": "artifact"}]

Transformation / operator

generate_independent_alternatives

Bounded public warrant

At portfolio, generate_independent_alternatives checks InventionCandidates before emitting Hypothesis Portfolio. Authority boundary: May propose and rank theories under frozen context; experiments constrain but never select a conceptual theory, and no proof is certified.

Conclusion

portfolio produces HypothesisPortfolio: Retain the quotient carrier as a formal-claim candidate.

Uncertainty

bounded by the declared artifact and open obligations

Assumptions

None recorded

Scope before / after / model / regime

{"after": "formal-claim candidate", "before": "conceptual candidate", "model": "competing carriers", "regime": "frozen theory context"}

Causal parents

["conceptual_frontier_deepening.base.portfolio.control"]

Artifact integrity status

integrity_valid

Scientific epistemic status

evidence_backed

Workflow disposition

accepted

Evidence links

evidence.926e08824a9582

Alternative / falsifier / failure links

alternative.c4c6ac46ac32a3, falsifier.e669f44b3c692a

Obligation links

None recorded

Artifact / receipt links

None recorded

Authority

{"absent": ["unscoped-epistemic-upgrade"], "exercised": ["portfolio-assessment"], "prohibitions_checked": ["no hidden scope promotion"]}

Handoff

None recorded

5.5 Inference 5: conceptual_frontier_deepening.base.reflect_discovery.inference

---

Trace disposition

Complete public dossier

Step / skill

reflect_discovery / reflective-conceptual-discovery

Question

At reflect_discovery, how does HypothesisPortfolio bear on ConceptualReflection? Which representation resolves the repeated bridge-object blocker under blind comparison?

Typed premises

[{"statement": "HypothesisPortfolio: The frozen context contains two non-dominated carrier constructions.", "type": "artifact"}]

Transformation / operator

revise_and_synthesize

Bounded public warrant

At reflect_discovery, revise_and_synthesize checks HypothesisPortfolio before emitting ConceptualReflection . Authority boundary: May propose and rank theories under frozen context; experiments constrain but never select a conceptual theory, and no proof is certified.

Conclusion

reflect_discovery produces ConceptualReflection: Retain the quotient carrier as a formal-claim candidate.

Uncertainty

bounded by the declared artifact and open obligations

Assumptions

None recorded

Scope before / after / model / regime

{"after": "formal-claim candidate", "before": "conceptual candidate", "model": "competing carriers", "regime": "frozen theory context"}

Causal parents

["conceptual_frontier_deepening.base.reflect_discovery.control"]

Artifact integrity status

integrity_valid

Scientific epistemic status

evidence_backed

Workflow disposition

accepted

Evidence links

evidence.d76d2ed1ef003a

Alternative / falsifier / failure links

alternative.ad231cd848a39e, falsifier.9544f55593c419

Obligation links

None recorded

Artifact / receipt links

None recorded

Authority

{"absent": ["unscoped-epistemic-upgrade"], "exercised": ["reflect_discovery-assessment"], "prohibitions_checked": ["no hidden scope promotion"]}

Handoff

None recorded

5.6 Inference 6: conceptual_frontier_deepening.base.blind_adjudicate.inference

---

Trace disposition

Complete public dossier

Step / skill

blind_adjudicate / blind-conceptual-preference-adjudicator

Question

At blind_adjudicate, how does TheoryCandidates bear on BlindAdjudication? Which representation resolves the repeated bridge-object blocker under blind comparison?

Typed premises

[{"statement": "TheoryCandidates: The frozen context contains two non-dominated carrier constructions.", "type": "artifact"}]

Transformation / operator

blind_multi_criterion_compare

Bounded public warrant

At blind_adjudicate, blind_multi_criterion_compare checks TheoryCandidates before emitting BlindAdjudication. Authority boundary: May propose and rank theories under frozen context; experiments constrain but never select a conceptual theory, and no proof is certified.

Conclusion

blind_adjudicate produces BlindAdjudication: Retain the quotient carrier as a formal-claim candidate.

Uncertainty

bounded by the declared artifact and open obligations

Assumptions

None recorded

Scope before / after / model / regime

{"after": "formal-claim candidate", "before": "conceptual candidate", "model": "competing carriers", "regime": "frozen theory context"}

Causal parents

["conceptual_frontier_deepening.base.blind_adjudicate.control"]

Artifact integrity status

integrity_valid

Scientific epistemic status

evidence_backed

Workflow disposition

accepted

Evidence links

evidence.8c094325baaa7d

Alternative / falsifier / failure links

alternative.6d66d838d3ebb2, falsifier.112f04a6ab9d1a

Obligation links

None recorded

Artifact / receipt links

None recorded

Authority

{"absent": ["unscoped-epistemic-upgrade"], "exercised": ["blind_adjudicate-assessment"], "prohibitions_checked": ["no hidden scope promotion"]}

Handoff

None recorded

5.7 Inference 7: conceptual_frontier_deepening.base.compile_atlas_defects.inference

---

Trace disposition

Complete public dossier

Step / skill

compile_atlas_defects / theory-atlas-defect-compiler

Question

At compile_atlas_defects, how does TheoryCandidates bear on TheoryAtlasDefects? Which representation resolves the repeated bridge-object blocker under blind comparison?

Typed premises

[{"statement": "TheoryCandidates: The frozen context contains two non-dominated carrier constructions.", "type": "artifact"}]

Transformation / operator

compute_result_atlas_delta

Bounded public warrant

At compile_atlas_defects, compute_result_atlas_delta checks TheoryCandidates before emitting TheoryAtlasDefects. Authority boundary: May propose and rank theories under frozen context; experiments constrain but never select a conceptual theory, and no proof is certified.

Conclusion

compile_atlas_defects produces TheoryAtlasDefects: Retain the quotient carrier as a formal-claim candidate.

Uncertainty

bounded by the declared artifact and open obligations

Assumptions

None recorded

Scope before / after / model / regime

{"after": "formal-claim candidate", "before": "conceptual candidate", "model": "competing carriers", "regime": "frozen theory context"}

Causal parents

["conceptual_frontier_deepening.base.compile_atlas_defects.control"]

Artifact integrity status

integrity_valid

Scientific epistemic status

evidence_backed

Workflow disposition

accepted

Evidence links

evidence.1dc9f21f646e13

Alternative / falsifier / failure links

alternative.fb8bd71c9dc1ec, falsifier.500c5788ed4318

Obligation links

None recorded

Artifact / receipt links

None recorded

Authority

{"absent": ["unscoped-epistemic-upgrade"], "exercised": ["compile_atlas_defects-assessment"], "prohibitions_checked": ["no hidden scope promotion"]}

Handoff

None recorded

5.8 Inference 8: conceptual_frontier_deepening.base.derive_central_routes.inference

---

Trace disposition

Complete public dossier

Step / skill

derive_central_routes / central-formal-conceptual-routes

Question

At derive_central_routes, how does BlindAdjudication bear on CentralConceptualRoutes? Which representation resolves the repeated bridge-object blocker under blind comparison?

Typed premises

[{"statement": "BlindAdjudication: The frozen context contains two non-dominated carrier constructions.", "type": "artifact"}]

Transformation / operator

artifact_grounded_transformation

Bounded public warrant

At derive_central_routes, artifact_grounded_transformation checks BlindAdjudication before emitting CentralConceptualRoutes. Authority boundary: May propose and rank theories under frozen context; experiments constrain but never select a conceptual theory, and no proof is certified.

Conclusion

derive_central_routes produces CentralConceptualRoutes: Retain the quotient carrier as a formal-claim candidate.

Uncertainty

bounded by the declared artifact and open obligations

Assumptions

None recorded

Scope before / after / model / regime

{"after": "formal-claim candidate", "before": "conceptual candidate", "model": "competing carriers", "regime": "frozen theory context"}

Causal parents

["conceptual_frontier_deepening.base.derive_central_routes.control"]

Artifact integrity status

integrity_valid

Scientific epistemic status

evidence_backed

Workflow disposition

accepted

Evidence links

evidence.da5b12b2f46ff6

Alternative / falsifier / failure links

alternative.6b211fde7c4a96, falsifier.434256ea44c6f3

Obligation links

None recorded

Artifact / receipt links

None recorded

Authority

{"absent": ["unscoped-epistemic-upgrade"], "exercised": ["derive_central_routes-assessment"], "prohibitions_checked": ["no hidden scope promotion"]}

Handoff

None recorded

5.9 Inference 9: conceptual_frontier_deepening.base.check_bridge_equivalence.inference

---

Trace disposition

Complete public dossier

Step / skill

check_bridge_equivalence / bridge-object-equivalence-checker

Question

At check_bridge_equivalence, how does CentralConceptualRoutes bear on BridgeEquivalenceDecision? Which representation resolves the repeated bridge-object blocker under blind comparison?

Typed premises

[{"statement": "CentralConceptualRoutes: The frozen context contains two non-dominated carrier constructi ons.", "type": "artifact"}]

Transformation / operator

test_semantic_equivalence

Bounded public warrant

At check_bridge_equivalence, test_semantic_equivalence checks CentralConceptualRoutes before emitting BridgeEquivalenceDecision. Authority boundary: May propose and rank theories under frozen context; experiments constrain but never select a conceptual theory, and no proof is certified.

Conclusion

check_bridge_equivalence produces BridgeEquivalenceDecision: Retain the quotient carrier as a formal-claim candidate.

Uncertainty

bounded by the declared artifact and open obligations

Assumptions

None recorded

Scope before / after / model / regime

{"after": "formal-claim candidate", "before": "conceptual candidate", "model": "competing carriers", "regime": "frozen theory context"}

Causal parents

["conceptual_frontier_deepening.base.check_bridge_equivalence.control"]

Artifact integrity status

integrity_valid

Scientific epistemic status

evidence_backed

Workflow disposition

accepted

Evidence links

evidence.75084779327c27

Alternative / falsifier / failure links

alternative.1b0e2eb3cb59d5, falsifier.a73d706a4560c5

Obligation links

None recorded

Artifact / receipt links

None recorded

Authority

{"absent": ["unscoped-epistemic-upgrade"], "exercised": ["check_bridge_equivalence-assessment"], "prohibitions_checked": ["no hidden scope promotion"]}

Handoff

None recorded

5.10 Inference 10: conceptual_frontier_deepening.base.produce_formal_claim.inference

Trace disposition

Complete public dossier

Step / skill

produce_formal_claim / formal-claim-producer

Question

At produce_formal_claim, how does CentralConceptualRoutes bear on FormalClaimProducerOutputV2? Which representation resolves the repeated bridge-object blocker under blind comparison?

Typed premises

[{"statement": "CentralConceptualRoutes: The frozen context contains two non-dominated carrier constructi ons.", "type": "artifact"}]

Transformation / operator

artifact_grounded_transformation

Bounded public warrant

At produce_formal_claim, artifact_grounded_transformation checks CentralConceptualRoutes before emitting FormalClaimProducerOutputV2. Authority boundary: May propose and rank theories under frozen context; experiments constrain but never select a conceptual theory, and no proof is certified.

Conclusion

produce_formal_claim produces FormalClaimProducerOutputV2: Retain the quotient carrier as a formal-claim candidate.

Uncertainty

bounded by the declared artifact and open obligations

Assumptions

None recorded

Scope before / after / model / regime

{"after": "formal-claim candidate", "before": "conceptual candidate", "model": "competing carriers", "regime": "frozen theory context"}

Causal parents

["conceptual_frontier_deepening.base.produce_formal_claim.control"]

Artifact integrity status

integrity_valid

Scientific epistemic status

evidence_backed

Workflow disposition

accepted

Evidence links

evidence.ecaa0a9bb2bf8c

Alternative / falsifier / failure links

alternative.8be5b4df9563bd, falsifier.1d9eb851906b70

Obligation links

None recorded

Artifact / receipt links

None recorded

Authority

{"absent": ["unscoped-epistemic-upgrade"], "exercised": ["produce_formal_claim-assessment"], "prohibitions_checked": ["no hidden scope promotion"]}

Handoff

None recorded

6 Alternatives, falsifiers, and failed paths

ID	Kind	Inference	Description	Disposition
alternative.08f9b00fdf4262	alternative	conceptual_frontier_deepening.base.plan_deepening.inference	Do not emit ConceptualDeepeningPlan: Blind adjudication rejects the enriched carrier because its equivalence warrant depends on the candidate label.	rejected
falsifier.4e30d6237d375c	falsifier	conceptual_frontier_deepening.base.plan_deepening.inference	A label-independent equivalence certificate reopens the enriched carrier.	Not recorded
alternative.09c22f7abfdc2d	alternative	conceptual_frontier_deepening.base.assimilate_literature.inference	Do not emit LiteratureFrontierMap: Blind adjudication rejects the enriched carrier because its equivalence warrant depends on the candidate label.	rejected
falsifier.f46d99fc154bf7	falsifier	conceptual_frontier_deepening.base.assimilate_literature.inference	A label-independent equivalence certificate reopens the enriched carrier.	Not recorded
alternative.a83038138ba717	alternative	conceptual_frontier_deepening.base.forge_invention.inference	Do not emit InventionCandidates: Blind adjudication rejects the enriched carrier because its equivalence warrant depends on the candidate label.	rejected
falsifier.a2eecd714d4e86	falsifier	conceptual_frontier_deepening.base.forge_invention.inference	A label-independent equivalence certificate reopens the enriched carrier.	Not recorded
alternative.c4c6ac46ac32a3	alternative	conceptual_frontier_deepening.base.portfolio.inference	Do not emit HypothesisPortfolio: Blind adjudication rejects the enriched carrier because its equivalence warrant depends on the candidate label.	rejected
falsifier.e669f44b3c692a	falsifier	conceptual_frontier_deepening.base.portfolio.inference	A label-independent equivalence certificate reopens the enriched carrier.	Not recorded
alternative.ad231cd848a39e	alternative	conceptual_frontier_deepening.base.reflect_discovery.inference	Do not emit ConceptualReflection: Blind adjudication rejects the enriched carrier because its equivalence warrant depends on the candidate label.	rejected

ID	Kind	Inference	Description	Disposition
falsifier.9544f55593c419	falsifier	conceptual_frontier_deepening.base.reflect_discovery.inference	A label-independent equivalence certificate reopens the enriched carrier.	Not recorded
alternative.3aae1d9365fda6	alternative	conceptual_frontier_deepening.base.synthesize.inference	Do not emit TheoryCandidates: Blind adjudication rejects the enriched carrier because its equivalence warrant depends on the candidate label.	accepted
falsifier.864b890ab6944a	falsifier	conceptual_frontier_deepening.base.synthesize.inference	A label-independent equivalence certificate reopens the enriched carrier.	Not recorded
alternatives_and_failure.f9eefbcbd73588be	failure	conceptual_frontier_deepening.base.synthesize.inference	synthesize rejects TheoryCandidates: Blind adjudication rejects the enriched carrier because its equivalence warrant depends on the candidate label.	Not recorded
alternative.6d66d838d3ebb2	alternative	conceptual_frontier_deepening.base.blind_adjudicate.inference	Do not emit BlindAdjudication: Blind adjudication rejects the enriched carrier because its equivalence warrant depends on the candidate label.	rejected
falsifier.112f04a6ab9d1a	falsifier	conceptual_frontier_deepening.base.blind_adjudicate.inference	A label-independent equivalence certificate reopens the enriched carrier.	Not recorded
alternative.fb8bd71c9dc1ec	alternative	conceptual_frontier_deepening.base.compile_atlas_defects.inference	Do not emit TheoryAtlasDefects: Blind adjudication rejects the enriched carrier because its equivalence warrant depends on the candidate label.	rejected
falsifier.500c5788ed4318	falsifier	conceptual_frontier_deepening.base.compile_atlas_defects.inference	A label-independent equivalence certificate reopens the enriched carrier.	Not recorded
alternative.6b211fde7c4a96	alternative	conceptual_frontier_deepening.base.derive_central_routes.inference	Do not emit CentralConceptualRoutes: Blind adjudication rejects the enriched carrier because its equivalence warrant depends on the candidate label.	rejected
falsifier.434256ea44c6f3	falsifier	conceptual_frontier_deepening.base.derive_central_routes.inference	A label-independent equivalence certificate reopens the enriched carrier.	Not recorded
alternative.1b0e2eb3cb59d5	alternative	conceptual_frontier_deepening.base.check_bridge_equivalence.inference	Do not emit BridgeEquivalenceDecision: Blind adjudication rejects the enriched carrier because its equivalence warrant depends on the candidate label.	rejected
falsifier.a73d706a4560c5	falsifier	conceptual_frontier_deepening.base.check_bridge_equivalence.inference	A label-independent equivalence certificate reopens the enriched carrier.	Not recorded

ID	Kind	Inference	Description	Disposition
alternative.8be5b4df9563bd	alternative	conceptual_frontier_deepening.base.produce_formal_claim.inference	Do not emit FormalClaimProducerOutputV2: Blind adjudication rejects the enriched carrier because its equivalence warrant depends on the candidate label.	rejected
falsifier.1d9eb851906b70	falsifier	conceptual_frontier_deepening.base.produce_formal_claim.inference	A label-independent equivalence certificate reopens the enriched carrier.	Not recorded

7 Cross-cutting validation ledgers

7.1 Scope transitions

Semantic ID	Complete recorded detail
conceptual_frontier_deepening.base.plan_deepening.inference	{"event_id": "conceptual_frontier_deepening.base.plan_deepening.inference", "scope": {"after": "formal-claim candidate", "before": "conceptual candidate", "model": "competing carriers", "regime": "frozen theory context"}, "step_id": "plan_deepening"}
conceptual_frontier_deepening.base.assimilate_literature.inference	{"event_id": "conceptual_frontier_deepening.base.assimilate_literature.inference", "scope": {"after": "formal-claim candidate", "before": "conceptual candidate", "model": "competing carriers", "regime": "frozen theory context"}, "step_id": "assimilate_literature"}
conceptual_frontier_deepening.base.forge_invention.inference	{"event_id": "conceptual_frontier_deepening.base.forge_invention.inference", "scope": {"after": "formal-claim candidate", "before": "conceptual candidate", "model": "competing carriers", "regime": "frozen theory context"}, "step_id": "forge_invention"}
conceptual_frontier_deepening.base.portfolio.inference	{"event_id": "conceptual_frontier_deepening.base.portfolio.inference", "scope": {"after": "formal-claim candidate", "before": "conceptual candidate", "model": "competing carriers", "regime": "frozen theory context"}, "step_id": "portfolio"}
conceptual_frontier_deepening.base.reflect_discovery.inference	{"event_id": "conceptual_frontier_deepening.base.reflect_discovery.inference", "scope": {"after": "formal-claim candidate", "before": "conceptual candidate", "model": "competing carriers", "regime": "frozen theory context"}, "step_id": "reflect_discovery"}
conceptual_frontier_deepening.base.synthesize.inference	{"event_id": "conceptual_frontier_deepening.base.synthesize.inference", "scope": {"after": "formal-claim candidate", "before": "conceptual candidate", "model": "competing carriers", "regime": "frozen theory context"}, "step_id": "synthesize"}
conceptual_frontier_deepening.base.blind_adjudicate.inference	{"event_id": "conceptual_frontier_deepening.base.blind_adjudicate.inference", "scope": {"after": "formal-claim candidate", "before": "conceptual candidate", "model": "competing carriers", "regime": "frozen theory context"}, "step_id": "blind_adjudicate"}
conceptual_frontier_deepening.base.compile_atlas_defects.inference	{"event_id": "conceptual_frontier_deepening.base.compile_atlas_defects.inference", "scope": {"after": "formal-claim candidate", "before": "conceptual candidate", "model": "competing carriers", "regime": "frozen theory context"}, "step_id": "compile_atlas_defects"}
conceptual_frontier_deepening.base.derive_central_routes.inference	{"event_id": "conceptual_frontier_deepening.base.derive_central_routes.inference", "scope": {"after": "formal-claim candidate", "before": "conceptual candidate", "model": "competing carriers", "regime": "frozen theory context"}, "step_id": "derive_central_routes"}
conceptual_frontier_deepening.base.check_bridge_equivalence.inference	{"event_id": "conceptual_frontier_deepening.base.check_bridge_equivalence.inference", "scope": {"after": "formal-claim candidate", "before": "conceptual candidate", "model": "competing carriers", "regime": "frozen theory context"}, "step_id": "check_bridge_equivalence"}

Semantic ID	Complete recorded detail
conceptual_frontier_deepening.base.produce_formal_claim.inference	{"event_id": "conceptual_frontier_deepening.base.produce_formal_claim.inference", "scope": {"after": "formal-claim candidate", "before": "conceptual candidate", "model": "competing carriers", "regime": "frozen theory context"}, "step_id": "produce_formal_claim"}

7.2 Authority boundaries

Semantic ID	Complete recorded detail
conceptual_frontier_deepening.base.plan_deepening.inference	{"absent": ["unscoped-epistemic-upgrade"], "event_id": "conceptual_frontier_deepening.base.plan_deepening.inference", "exercised": ["plan_deepening-assessment"], "prohibitions_checked": ["no hidden scope promotion"], "step_id": "plan_deepening"}
conceptual_frontier_deepening.base.assimilate_literature.inference	{"absent": ["unscoped-epistemic-upgrade"], "event_id": "conceptual_frontier_deepening.base.assimilate_literature.inference", "exercised": ["assimilate_literature-assessment"], "prohibitions_checked": ["no hidden scope promotion"], "step_id": "assimilate_literature"}
conceptual_frontier_deepening.base.forge_invention.inference	{"absent": ["unscoped-epistemic-upgrade"], "event_id": "conceptual_frontier_deepening.base.forge_invention.inference", "exercised": ["forge_invention-assessment"], "prohibitions_checked": ["no hidden scope promotion"], "step_id": "forge_invention"}
conceptual_frontier_deepening.base.portfolio.inference	{"absent": ["unscoped-epistemic-upgrade"], "event_id": "conceptual_frontier_deepening.base.portfolio.inference", "exercised": ["portfolio-assessment"], "prohibitions_checked": ["no hidden scope promotion"], "step_id": "portfolio"}
conceptual_frontier_deepening.base.reflect_discovery.inference	{"absent": ["unscoped-epistemic-upgrade"], "event_id": "conceptual_frontier_deepening.base.reflect_discovery.inference", "exercised": ["reflect_discovery-assessment"], "prohibitions_checked": ["no hidden scope promotion"], "step_id": "reflect_discovery"}
conceptual_frontier_deepening.base.synthesize.inference	{"absent": ["unscoped-epistemic-upgrade"], "event_id": "conceptual_frontier_deepening.base.synthesize.inference", "exercised": ["synthesize-assessment"], "prohibitions_checked": ["no hidden scope promotion"], "step_id": "synthesize"}
conceptual_frontier_deepening.base.blind_adjudicate.inference	{"absent": ["unscoped-epistemic-upgrade"], "event_id": "conceptual_frontier_deepening.base.blind_adjudicate.inference", "exercised": ["blind_adjudicate-assessment"], "prohibitions_checked": ["no hidden scope promotion"], "step_id": "blind_adjudicate"}
conceptual_frontier_deepening.base.compile_atlas_defects.inference	{"absent": ["unscoped-epistemic-upgrade"], "event_id": "conceptual_frontier_deepening.base.compile_atlas_defects.inference", "exercised": ["compile_atlas_defects-assessment"], "prohibitions_checked": ["no hidden scope promotion"], "step_id": "compile_atlas_defects"}
conceptual_frontier_deepening.base.derive_central_routes.inference	{"absent": ["unscoped-epistemic-upgrade"], "event_id": "conceptual_frontier_deepening.base.derive_central_routes.inference", "exercised": ["derive_central_routes-assessment"], "prohibitions_checked": ["no hidden scope promotion"], "step_id": "derive_central_routes"}
conceptual_frontier_deepening.base.check_bridge_equivalence.inference	{"absent": ["unscoped-epistemic-upgrade"], "event_id": "conceptual_frontier_deepening.base.check_bridge_equivalence.inference", "exercised": ["check_bridge_equivalence-assessment"], "prohibitions_checked": ["no hidden scope promotion"], "step_id": "check_bridge_equivalence"}
conceptual_frontier_deepening.base.produce_formal_claim.inference	{"absent": ["unscoped-epistemic-upgrade"], "event_id": "conceptual_frontier_deepening.base.produce_formal_claim.inference", "exercised": ["produce_formal_claim-assessment"], "prohibitions_checked": ["no hidden scope promotion"], "step_id": "produce_formal_claim"}

7.3 Open and discharged obligations

Semantic ID	Complete recorded detail
obligation.03f98d493ed190	{"event_id": "conceptual_frontier_deepening.base.synthesize.inference", "obligation_id": "obligation.03f98d493ed190", "requirement": "A label-independent equivalence certificate reopens the enriched carrier.", "status": "open"}

7.4 Handoffs

Semantic ID	Complete recorded detail
conceptual_frontier_deepening.handoff	{"accepted": true, "destination_workflow_id": "theorem_proof_delivery", "event_id": "conceptual_frontier_deepening.handoff"}

A Complete chronological lifecycle ledger

This ledger records execution order. A precedes relation is not evidential support or entailment.

Event ID	Seq.	Family	Step	Disposition	Parents
conceptual_frontier_deepening.base.plan_deepening.control	1	authorization	plan_deepening	authorized	None
conceptual_frontier_deepening.base.plan_deepening.inference	2	inference	plan_deepening	accepted	["conceptual_frontier_deepening.base.plan_deepening.control"]
conceptual_frontier_deepening.base.assimilate_literature.control	3	authorization	assimilate_literature	authorized	["conceptual_frontier_deepening.base.plan_deepening.inference"]
conceptual_frontier_deepening.base.assimilate_literature.inference	4	inference	assimilate_literature	accepted	["conceptual_frontier_deepening.base.assimilate_literature.control"]
conceptual_frontier_deepening.base.forge_invention.control	5	authorization	forge_invention	authorized	["conceptual_frontier_deepening.base.assimilate_literature.inference"]
conceptual_frontier_deepening.base.forge_invention.inference	6	inference	forge_invention	accepted	["conceptual_frontier_deepening.base.forge_invention.control"]
conceptual_frontier_deepening.base.portfolio.control	7	authorization	portfolio	authorized	["conceptual_frontier_deepening.base.forge_invention.inference"]
conceptual_frontier_deepening.base.portfolio.inference	8	inference	portfolio	accepted	["conceptual_frontier_deepening.base.portfolio.control"]
conceptual_frontier_deepening.base.reflect_discovery.control	9	authorization	reflect_discovery	authorized	["conceptual_frontier_deepening.base.portfolio.inference"]
conceptual_frontier_deepening.base.reflect_discovery.inference	10	inference	reflect_discovery	accepted	["conceptual_frontier_deepening.base.reflect_discovery.control"]

Event ID	Seq.	Family	Step	Disposition	Parents
conceptual_frontier_deepening.base.synthesize.control	11	authorization	synthesize	authorized	["conceptual_frontier_deepening.base.reflect_discovery.inference"]
conceptual_frontier_deepening.base.synthesize.inference	12	failure	synthesize	failed	["conceptual_frontier_deepening.base.synthesize.control"]
conceptual_frontier_deepening.base.blind_adjudicate.control	13	authorization	blind_adjudicate	authorized	["conceptual_frontier_deepening.base.synthesize.inference"]
conceptual_frontier_deepening.base.blind_adjudicate.inference	14	inference	blind_adjudicate	accepted	["conceptual_frontier_deepening.base.blind_adjudicate.control"]
conceptual_frontier_deepening.base.compile_atlas_defects.control	15	authorization	compile_atlas_defects	authorized	["conceptual_frontier_deepening.base.blind_adjudicate.inference"]
conceptual_frontier_deepening.base.compile_atlas_defects.inference	16	inference	compile_atlas_defects	accepted	["conceptual_frontier_deepening.base.compile_atlas_defects.control"]
conceptual_frontier_deepening.base.derive_central_routes.control	17	authorization	derive_central_routes	authorized	["conceptual_frontier_deepening.base.compile_atlas_defects.inference"]
conceptual_frontier_deepening.base.derive_central_routes.inference	18	inference	derive_central_routes	accepted	["conceptual_frontier_deepening.base.derive_central_routes.control"]
conceptual_frontier_deepening.base.check_bridge_equivalence.control	19	authorization	check_bridge_equivalence	authorized	["conceptual_frontier_deepening.base.derive_central_routes.inference"]
conceptual_frontier_deepening.base.check_bridge_equivalence.inference	20	inference	check_bridge_equivalence	accepted	["conceptual_frontier_deepening.base.check_bridge_equivalence.control"]
conceptual_frontier_deepening.base.produce_formal_claim.control	21	authorization	produce_formal_claim	authorized	["conceptual_frontier_deepening.base.check_bridge_equivalence.inference"]
conceptual_frontier_deepening.base.produce_formal_claim.inference	22	inference	produce_formal_claim	accepted	["conceptual_frontier_deepening.base.produce_formal_claim.control"]
conceptual_frontier_deepening.handoff	23	handoff	Not recorded	handed_off	["conceptual_frontier_deepening.base.produce_formal_claim.inference"]

B Claim ledger

Semantic ID	Complete recorded detail
conceptual_frontier_deepening.base.plan_deepening.inference	{"claim_id": "conceptual_frontier_deepening.base.plan_deepening.inference", "epistemic_status": "evidence_backed", "event_id": "conceptual_frontier_deepening.base.plan_deepening.inference", "evidence_ids": ["evidence.d20af4ce1f3d92"], "integrity_status": "integrity_valid", "obligation_ids": [], "scope": {"after": "formal-claim candidate", "before": "conceptual candidate", "model": "competing carriers", "regime": "frozen theory context"}, "text": "plan_deepening produces ConceptualDeepeningPlan: Retain the quotient carrier as a formal-claim candidate.", "trace_incomplete": false, "workflow_disposition": "accepted"}
conceptual_frontier_deepening.base.assimilate_literature.inference	{"claim_id": "conceptual_frontier_deepening.base.assimilate_literature.inference", "epistemic_status": "evidence_backed", "event_id": "conceptual_frontier_deepening.base.assimilate_literature.inference", "evidence_ids": ["evidence.0697c302322c1f"], "integrity_status": "integrity_valid", "obligation_ids": [], "scope": {"after": "formal-claim candidate", "before": "conceptual candidate", "model": "competing carriers", "regime": "frozen theory context"}, "text": "assimilate_literature produces LiteratureFrontierMap: Retain the quotient carrier as a formal-claim candidate.", "trace_incomplete": false, "workflow_disposition": "accepted"}
conceptual_frontier_deepening.base.forge_invention.inference	{"claim_id": "conceptual_frontier_deepening.base.forge_invention.inference", "epistemic_status": "evidence_backed", "event_id": "conceptual_frontier_deepening.base.forge_invention.inference", "evidence_ids": ["evidence.c80c8c4a3127a7"], "integrity_status": "integrity_valid", "obligation_ids": [], "scope": {"after": "formal-claim candidate", "before": "conceptual candidate", "model": "competing carriers", "regime": "frozen theory context"}, "text": "forge_invention produces InventionCandidates: Retain the quotient carrier as a formal-claim candidate.", "trace_incomplete": false, "workflow_disposition": "accepted"}
conceptual_frontier_deepening.base.portfolio.inference	{"claim_id": "conceptual_frontier_deepening.base.portfolio.inference", "epistemic_status": "evidence_backed", "event_id": "conceptual_frontier_deepening.base.portfolio.inference", "evidence_ids": ["evidence.926e08824a9582"], "integrity_status": "integrity_valid", "obligation_ids": [], "scope": {"after": "formal-claim candidate", "before": "conceptual candidate", "model": "competing carriers", "regime": "frozen theory context"}, "text": "portfolio produces HypothesisPortfolio: Retain the quotient carrier as a formal-claim candidate.", "trace_incomplete": false, "workflow_disposition": "accepted"}
conceptual_frontier_deepening.base.reflect_discovery.inference	{"claim_id": "conceptual_frontier_deepening.base.reflect_discovery.inference", "epistemic_status": "evidence_backed", "event_id": "conceptual_frontier_deepening.base.reflect_discovery.inference", "evidence_ids": ["evidence.d76d2ed1ef003a"], "integrity_status": "integrity_valid", "obligation_ids": [], "scope": {"after": "formal-claim candidate", "before": "conceptual candidate", "model": "competing carriers", "regime": "frozen theory context"}, "text": "reflect_discovery produces ConceptualReflection: Retain the quotient carrier as a formal-claim candidate.", "trace_incomplete": false, "workflow_disposition": "accepted"}
conceptual_frontier_deepening.base.synthesize.inference	{"claim_id": "conceptual_frontier_deepening.base.synthesize.inference", "epistemic_status": "quarantined", "event_id": "conceptual_frontier_deepening.base.synthesize.inference", "evidence_ids": ["evidence.d9200d4125b81c"], "integrity_status": "integrity_valid", "obligation_ids": ["obligation.03f98d493ed190"], "scope": {"after": "formal-claim candidate", "before": "conceptual candidate", "model": "competing carriers", "regime": "frozen theory context"}, "text": "synthesize rejects TheoryCandidates: Blind adjudication rejects the enriched carrier because its equivalence warrant depends on the candidate label.", "trace_incomplete": false, "workflow_disposition": "failed"}
conceptual_frontier_deepening.base.blind_adjudicate.inference	{"claim_id": "conceptual_frontier_deepening.base.blind_adjudicate.inference", "epistemic_status": "evidence_backed", "event_id": "conceptual_frontier_deepening.base.blind_adjudicate.inference", "evidence_ids": ["evidence.8c094325baaa7d"], "integrity_status": "integrity_valid", "obligation_ids": [], "scope": {"after": "formal-claim candidate", "before": "conceptual candidate", "model": "competing carriers", "regime": "frozen theory context"}, "text": "blind_adjudicate produces BlindAdjudication: Retain the quotient carrier as a formal-claim candidate.", "trace_incomplete": false, "workflow_disposition": "accepted"}

Semantic ID	Complete recorded detail
conceptual_frontier_deepening.base.compile_atlas_defects.inference	{"claim_id": "conceptual_frontier_deepening.base.compile_atlas_defects.inference", "epistemic_status": "evidence_backed", "event_id": "conceptual_frontier_deepening.base.compile_atlas_defects.inference", "evidence_ids": ["evidence.1dc9f21f646e13"], "integrity_status": "integrity_valid", "obligation_ids": [], "scope": {"after": "formal-claim candidate", "before": "conceptual candidate", "model": "competing carriers", "regime": "frozen theory context"}, "text": "compile_atlas_defects produces TheoryAtlasDefects: Retain the quotient carrier as a formal-claim candidate.", "trace_incomplete": false, "workflow_disposition": "accepted"}
conceptual_frontier_deepening.base.derive_central_routes.inference	{"claim_id": "conceptual_frontier_deepening.base.derive_central_routes.inference", "epistemic_status": "evidence_backed", "event_id": "conceptual_frontier_deepening.base.derive_central_routes.inference", "evidence_ids": ["evidence.da5b12b2f46ff6"], "integrity_status": "integrity_valid", "obligation_ids": [], "scope": {"after": "formal-claim candidate", "before": "conceptual candidate", "model": "competing carriers", "regime": "frozen theory context"}, "text": "derive_central_routes produces CentralConceptualRoutes: Retain the quotient carrier as a formal-claim candidate.", "trace_incomplete": false, "workflow_disposition": "accepted"}
conceptual_frontier_deepening.base.check_bridge_equivalence.inference	{"claim_id": "conceptual_frontier_deepening.base.check_bridge_equivalence.inference", "epistemic_status": "evidence_backed", "event_id": "conceptual_frontier_deepening.base.check_bridge_equivalence.inference", "evidence_ids": ["evidence.75084779327c27"], "integrity_status": "integrity_valid", "obligation_ids": [], "scope": {"after": "formal-claim candidate", "before": "conceptual candidate", "model": "competing carriers", "regime": "frozen theory context"}, "text": "check_bridge_equivalence produces BridgeEquivalenceDecision: Retain the quotient carrier as a formal-claim candidate.", "trace_incomplete": false, "workflow_disposition": "accepted"}
conceptual_frontier_deepening.base.produce_formal_claim.inference	{"claim_id": "conceptual_frontier_deepening.base.produce_formal_claim.inference", "epistemic_status": "evidence_backed", "event_id": "conceptual_frontier_deepening.base.produce_formal_claim.inference", "evidence_ids": ["evidence.ecaa0a9bb2bf8c"], "integrity_status": "integrity_valid", "obligation_ids": [], "scope": {"after": "formal-claim candidate", "before": "conceptual candidate", "model": "competing carriers", "regime": "frozen theory context"}, "text": "produce_formal_claim produces FormalClaimProducerOutputV2: Retain the quotient carrier as a formal-claim candidate.", "trace_incomplete": false, "workflow_disposition": "accepted"}

C Typed evidence ledger

Semantic ID	Complete recorded detail
evidence.d20af4ce1f3d92	{"artifact_type": "FrozenDiscoveryContext", "digest": "61c8ddbab2b6d4dded3cbe6cf9348c5fd6c658796985c80675774399043f4a90", "event_id": "conceptual_frontier_deepening.base.plan_deepening.inference", "evidence_id": "evidence.d20af4ce1f3d92", "path": "tests/fixtures/reasoning_trace/evidence/workflow-evidence.json", "relation": "supports"}
evidence.0697c302322c1f	{"artifact_type": "FrozenDiscoveryContext", "digest": "61c8ddbab2b6d4dded3cbe6cf9348c5fd6c658796985c80675774399043f4a90", "event_id": "conceptual_frontier_deepening.base.assimilate_literature.inference", "evidence_id": "evidence.0697c302322c1f", "path": "tests/fixtures/reasoning_trace/evidence/workflow-evidence.json", "relation": "supports"}
evidence.c80c8c4a3127a7	{"artifact_type": "LiteratureFrontierMap", "digest": "61c8ddbab2b6d4dded3cbe6cf9348c5fd6c658796985c80675774399043f4a90", "event_id": "conceptual_frontier_deepening.base.forge_invention.inference", "evidence_id": "evidence.c80c8c4a3127a7", "path": "tests/fixtures/reasoning_trace/evidence/workflow-evidence.json", "relation": "supports"}
evidence.926e08824a9582	{"artifact_type": "InventionCandidates", "digest": "61c8ddbab2b6d4dded3cbe6cf9348c5fd6c658796985c80675774399043f4a90", "event_id": "conceptual_frontier_deepening.base.portfolio.inference", "evidence_id": "evidence.926e08824a9582", "path": "tests/fixtures/reasoning_trace/evidence/workflow-evidence.json", "relation": "supports"}

Semantic ID	Complete recorded detail
evidence.d76d2ed1ef003a	{"artifact_type": "HypothesisPortfolio", "digest": "61c8ddbab2b6d4dded3cbe6cf9348c5fd6c658796985c80675774399043f4a90", "event_id": "conceptual_frontier_deepening.base.reflect_discovery.inference", "evidence_id": "evidence.d76d2ed1ef003a", "path": "tests/fixtures/reasoning_trace/evidence/workflow-evidence.json", "relation": "supports"}
evidence.d9200d4125b81c	{"artifact_type": "ConceptualReflection", "digest": "61c8ddbab2b6d4dded3cbe6cf9348c5fd6c658796985c80675774399043f4a90", "event_id": "conceptual_frontier_deepening.base.synthesize.inference", "evidence_id": "evidence.d9200d4125b81c", "path": "tests/fixtures/reasoning_trace/evidence/workflow-evidence.json", "relation": "refutes"}
evidence.8c094325baaa7d	{"artifact_type": "TheoryCandidates", "digest": "61c8ddbab2b6d4dded3cbe6cf9348c5fd6c658796985c80675774399043f4a90", "event_id": "conceptual_frontier_deepening.base.blind_adjudicate.inference", "evidence_id": "evidence.8c094325baaa7d", "path": "tests/fixtures/reasoning_trace/evidence/workflow-evidence.json", "relation": "supports"}
evidence.1dc9f21f646e13	{"artifact_type": "TheoryCandidates", "digest": "61c8ddbab2b6d4dded3cbe6cf9348c5fd6c658796985c80675774399043f4a90", "event_id": "conceptual_frontier_deepening.base.compile_atlas_defects.inference", "evidence_id": "evidence.1dc9f21f646e13", "path": "tests/fixtures/reasoning_trace/evidence/workflow-evidence.json", "relation": "supports"}
evidence.da5b12b2f46ff6	{"artifact_type": "BlindAdjudication", "digest": "61c8ddbab2b6d4dded3cbe6cf9348c5fd6c658796985c80675774399043f4a90", "event_id": "conceptual_frontier_deepening.base.derive_central_routes.inference", "evidence_id": "evidence.da5b12b2f46ff6", "path": "tests/fixtures/reasoning_trace/evidence/workflow-evidence.json", "relation": "supports"}
evidence.75084779327c27	{"artifact_type": "CentralConceptualRoutes", "digest": "61c8ddbab2b6d4dded3cbe6cf9348c5fd6c658796985c80675774399043f4a90", "event_id": "conceptual_frontier_deepening.base.check_bridge_equivalence.inference", "evidence_id": "evidence.75084779327c27", "path": "tests/fixtures/reasoning_trace/evidence/workflow-evidence.json", "relation": "supports"}
evidence.ecaa0a9bb2bf8c	{"artifact_type": "CentralConceptualRoutes", "digest": "61c8ddbab2b6d4dded3cbe6cf9348c5fd6c658796985c80675774399043f4a90", "event_id": "conceptual_frontier_deepening.base.produce_formal_claim.inference", "evidence_id": "evidence.ecaa0a9bb2bf8c", "path": "tests/fixtures/reasoning_trace/evidence/workflow-evidence.json", "relation": "supports"}

D Decision and gate ledger

Semantic ID	Complete recorded detail
conceptual_frontier_deepening.base.plan_deepening.control	{"disposition": "authorized", "event_id": "conceptual_frontier_deepening.base.plan_deepening.control", "family": "authorization", "reason": "", "step_id": "plan_deepening"}
conceptual_frontier_deepening.base.assimilate_literature.control	{"disposition": "authorized", "event_id": "conceptual_frontier_deepening.base.assimilate_literature.control", "family": "authorization", "reason": "", "step_id": "assimilate_literature"}
conceptual_frontier_deepening.base.forge_invention.control	{"disposition": "authorized", "event_id": "conceptual_frontier_deepening.base.forge_invention.control", "family": "authorization", "reason": "", "step_id": "forge_invention"}
conceptual_frontier_deepening.base.portfolio.control	{"disposition": "authorized", "event_id": "conceptual_frontier_deepening.base.portfolio.control", "family": "authorization", "reason": "", "step_id": "portfolio"}
conceptual_frontier_deepening.base.reflect_discovery.control	{"disposition": "authorized", "event_id": "conceptual_frontier_deepening.base.reflect_discovery.control", "family": "authorization", "reason": "", "step_id": "reflect_discovery"}

Semantic ID	Complete recorded detail
conceptual_frontier_deepening.base.synthesize.control	{"disposition": "authorized", "event_id": "conceptual_frontier_deepening.base.synthesize.control", "family": "authorization", "reason": "", "step_id": "synthesize"}
conceptual_frontier_deepening.base.blind_adjudicate.control	{"disposition": "authorized", "event_id": "conceptual_frontier_deepening.base.blind_adjudicate.control", "family": "authorization", "reason": "", "step_id": "blind_adjudicate"}
conceptual_frontier_deepening.base.compile_atlas_defects.control	{"disposition": "authorized", "event_id": "conceptual_frontier_deepening.base.compile_atlas_defects.control", "family": "authorization", "reason": "", "step_id": "compile_atlas_defects"}
conceptual_frontier_deepening.base.derive_central_routes.control	{"disposition": "authorized", "event_id": "conceptual_frontier_deepening.base.derive_central_routes.control", "family": "authorization", "reason": "", "step_id": "derive_central_routes"}
conceptual_frontier_deepening.base.check_bridge_equivalence.control	{"disposition": "authorized", "event_id": "conceptual_frontier_deepening.base.check_bridge_equivalence.control", "family": "authorization", "reason": "", "step_id": "check_bridge_equivalence"}
conceptual_frontier_deepening.base.produce_formal_claim.control	{"disposition": "authorized", "event_id": "conceptual_frontier_deepening.base.produce_formal_claim.control", "family": "authorization", "reason": "", "step_id": "produce_formal_claim"}

E Artifact and receipt ledger

Semantic ID	Complete recorded detail
No artifact and receipt ledger recorded.	

F Specialized projection ledger

Semantic ID	Complete recorded detail
No specialized projection ledger recorded.	

G Causal and workflow graph ledger

Typed edge summary

precedes: 32, contains: 23, supports: 10, refutes: 1, produces: 1

Edge meanings are explicit: precedes is chronology; supports/refutes/constrains are epistemic relations; contains/produces/branches/joins/refines/validates/blocks/hands-off are workflow relations.

G.1 Nodes

Node	Type	Label	Status
conceptual_frontier_deepening.base.plan_deepening.control	event	authorization	Not recorded
conceptual_frontier_deepening.base.plan_deepening.inference	event	plan_deepening produces Conceptual DeepeningPlan: Retain the quotient carrier as a formal-claim candidate.	Not recorded

Node	Type	Label	Status
evidence.d20af4ce1f3d92	evidence	tests/fixtures/reasoning_trace/evidence/workflow-evidence.json	Not recorded
conceptual_frontier_deepening.base.assimilate_literature.control	event	authorization	Not recorded
conceptual_frontier_deepening.base.assimilate_literature.inference	event	assimilate_literature produces Literature FrontierMap: Retain the quotient carrier as a formal-claim candidate.	Not recorded
evidence.0697c302322c1f	evidence	tests/fixtures/reasoning_trace/evidence/workflow-evidence.json	Not recorded
conceptual_frontier_deepening.base.forge_invention.control	event	authorization	Not recorded
conceptual_frontier_deepening.base.forge_invention.inference	event	forge_invention produces InventionCandidates: Retain the quotient carrier as a formal-claim candidate.	Not recorded
evidence.c80c8c4a3127a7	evidence	tests/fixtures/reasoning_trace/evidence/workflow-evidence.json	Not recorded
conceptual_frontier_deepening.base.portfolio.control	event	authorization	Not recorded
conceptual_frontier_deepening.base.portfolio.inference	event	portfolio produces HypothesisPortfolio: Retain the quotient carrier as a formal-claim candidate.	Not recorded
evidence.926e08824a9582	evidence	tests/fixtures/reasoning_trace/evidence/workflow-evidence.json	Not recorded
conceptual_frontier_deepening.base.reflect_discovery.control	event	authorization	Not recorded
conceptual_frontier_deepening.base.reflect_discovery.inference	event	reflect_discovery produces Conceptual Reflection: Retain the quotient carrier as a formal-claim candidate.	Not recorded
evidence.d76d2ed1ef003a	evidence	tests/fixtures/reasoning_trace/evidence/workflow-evidence.json	Not recorded
conceptual_frontier_deepening.base.synthesize.control	event	authorization	Not recorded
conceptual_frontier_deepening.base.synthesize.inference	event	synthesize rejects TheoryCandidates: Blind adjudication rejects the enriched carrier because its equivalence warrant depends on the candidate label.	Not recorded
evidence.d9200d4125b81c	evidence	tests/fixtures/reasoning_trace/evidence/workflow-evidence.json	Not recorded
obligation.03f98d493ed190	obligation	A label-independent equivalence certificate reopens the enriched carrier.	Not recorded
conceptual_frontier_deepening.base.blind_adjudicate.control	event	authorization	Not recorded
conceptual_frontier_deepening.base.blind_adjudicate.inference	event	blind_adjudicate produces BlindAdjudication: Retain the quotient carrier as a formal-claim candidate.	Not recorded
evidence.8c094325baaa7d	evidence	tests/fixtures/reasoning_trace/evidence/workflow-evidence.json	Not recorded
conceptual_frontier_deepening.base.compile_atlas_defects.control	event	authorization	Not recorded
conceptual_frontier_deepening.base.compile_atlas_defects.inference	event	compile_atlas_defects produces TheoryAtlasDefects: Retain the quotient carrier as a formal-claim candidate.	Not recorded
evidence.1dc9f21f646e13	evidence	tests/fixtures/reasoning_trace/evidence/workflow-evidence.json	Not recorded
conceptual_frontier_deepening.base.derive_central_routes.control	event	authorization	Not recorded

Node	Type	Label	Status
conceptual_frontier_deepening.base.derive_central_routes.inference	event	derive_central_routes produces CentralConceptualRoutes: Retain the quotient carrier as a formal-claim candidate.	Not recorded
evidence.da5b12b2f46ff6	evidence	tests/fixtures/reasoning_trace/evidence/workflow-evidence.json	Not recorded
conceptual_frontier_deepening.base.check_bridge_equivalence.control	event	authorization	Not recorded
conceptual_frontier_deepening.base.check_bridge_equivalence.inference	event	check_bridge_equivalence produces BridgeEquivalenceDecision: Retain the quotient carrier as a formal-claim candidate.	Not recorded
evidence.75084779327c27	evidence	tests/fixtures/reasoning_trace/evidence/workflow-evidence.json	Not recorded
conceptual_frontier_deepening.base.produce_formal_claim.control	event	authorization	Not recorded
conceptual_frontier_deepening.base.produce_formal_claim.inference	event	produce_formal_claim produces FormalClaimProducerOutputV2: Retain the quotient carrier as a formal-claim candidate.	Not recorded
evidence.ecaa0a9bb2bf8c	evidence	tests/fixtures/reasoning_trace/evidence/workflow-evidence.json	Not recorded
conceptual_frontier_deepening.handoff	event	Typed handoff to theorem_proof_delivery	Not recorded
step:plan_deepening	step	plan_deepening	observed
step:assimilate_literature	step	assimilate_literature	observed
step:forge_invention	step	forge_invention	observed
step:portfolio	step	portfolio	observed
step:reflect_discovery	step	reflect_discovery	observed
step:synthesize	step	synthesize	observed
step:blind_adjudicate	step	blind_adjudicate	observed
step:compile_atlas_defects	step	compile_atlas_defects	observed
step:derive_central_routes	step	derive_central_routes	observed
step:check_bridge_equivalence	step	check_bridge_equivalence	observed
step:produce_formal_claim	step	produce_formal_claim	observed

G.2 Complete typed edge table

Source	Relation	Target	Adapter / detail
conceptual_frontier_deepening.base.plan_deepening.control	precedes	conceptual_frontier_deepening.base.plan_deepening.inference	None
evidence.d20af4ce1f3d92	supports	conceptual_frontier_deepening.base.plan_deepening.inference	None
conceptual_frontier_deepening.base.plan_deepening.inference	precedes	conceptual_frontier_deepening.base.assimilate_literature.control	None
conceptual_frontier_deepening.base.assimilate_literature.control	precedes	conceptual_frontier_deepening.base.assimilate_literature.inference	None
evidence.0697c302322c1f	supports	conceptual_frontier_deepening.base.assimilate_literature.inference	None
conceptual_frontier_deepening.base.assimilate_literature.inference	precedes	conceptual_frontier_deepening.base.forge_invention.control	None

Source	Relation	Target	Adapter / detail
conceptual_frontier_deepening.base.forge_invention.control	precedes	conceptual_frontier_deepening.base.forge_invention.inference	None
evidence.c80c8c4a3127a7	supports	conceptual_frontier_deepening.base.forge_invention.inference	None
conceptual_frontier_deepening.base.forge_invention.inference	precedes	conceptual_frontier_deepening.base.portfolio.control	None
conceptual_frontier_deepening.base.portfolio.control	precedes	conceptual_frontier_deepening.base.portfolio.inference	None
evidence.926e08824a9582	supports	conceptual_frontier_deepening.base.portfolio.inference	None
conceptual_frontier_deepening.base.portfolio.inference	precedes	conceptual_frontier_deepening.base.reflect_discovery.control	None
conceptual_frontier_deepening.base.reflect_discovery.control	precedes	conceptual_frontier_deepening.base.reflect_discovery.inference	None
evidence.d76d2ed1ef003a	supports	conceptual_frontier_deepening.base.reflect_discovery.inference	None
conceptual_frontier_deepening.base.reflect_discovery.inference	precedes	conceptual_frontier_deepening.base.synthesize.control	None
conceptual_frontier_deepening.base.synthesize.control	precedes	conceptual_frontier_deepening.base.synthesize.inference	None
evidence.d9200d4125b81c	refutes	conceptual_frontier_deepening.base.synthesize.inference	None
conceptual_frontier_deepening.base.synthesize.inference	produces	obligation.03f98d493ed190	None
conceptual_frontier_deepening.base.synthesize.inference	precedes	conceptual_frontier_deepening.base.blind_adjudicate.control	None
conceptual_frontier_deepening.base.blind_adjudicate.control	precedes	conceptual_frontier_deepening.base.blind_adjudicate.inference	None
evidence.8c094325baaa7d	supports	conceptual_frontier_deepening.base.blind_adjudicate.inference	None
conceptual_frontier_deepening.base.blind_adjudicate.inference	precedes	conceptual_frontier_deepening.base.compile_atlas_defects.control	None
conceptual_frontier_deepening.base.compile_atlas_defects.control	precedes	conceptual_frontier_deepening.base.compile_atlas_defects.inference	None
evidence.1dc9f21f646e13	supports	conceptual_frontier_deepening.base.compile_atlas_defects.inference	None
conceptual_frontier_deepening.base.compile_atlas_defects.inference	precedes	conceptual_frontier_deepening.base.derive_central_routes.control	None
conceptual_frontier_deepening.base.derive_central_routes.control	precedes	conceptual_frontier_deepening.base.derive_central_routes.inference	None
evidence.da5b12b2f46ff6	supports	conceptual_frontier_deepening.base.derive_central_routes.inference	None

Source	Relation	Target	Adapter / detail
conceptual_frontier_deepening.base.derive_central_routes.inference	precedes	conceptual_frontier_deepening.base.check_bridge_equivalence.control	None
conceptual_frontier_deepening.base.check_bridge_equivalence.control	precedes	conceptual_frontier_deepening.base.check_bridge_equivalence.inference	None
evidence.75084779327c27	supports	conceptual_frontier_deepening.base.check_bridge_equivalence.inference	None
conceptual_frontier_deepening.base.check_bridge_equivalence.inference	precedes	conceptual_frontier_deepening.base.produce_formal_claim.control	None
conceptual_frontier_deepening.base.produce_formal_claim.control	precedes	conceptual_frontier_deepening.base.produce_formal_claim.inference	None
evidence.ecaa0a9bb2bf8c	supports	conceptual_frontier_deepening.base.produce_formal_claim.inference	None
conceptual_frontier_deepening.base.produce_formal_claim.inference	precedes	conceptual_frontier_deepening.handoff	None
step:plan_deepening	contains	conceptual_frontier_deepening.base.plan_deepening.control	None
step:plan_deepening	contains	conceptual_frontier_deepening.base.plan_deepening.inference	None
step:assimilate_literature	contains	conceptual_frontier_deepening.base.assimilate_literature.control	None
step:assimilate_literature	contains	conceptual_frontier_deepening.base.assimilate_literature.inference	None
step:forge_invention	contains	conceptual_frontier_deepening.base.forge_invention.control	None
step:forge_invention	contains	conceptual_frontier_deepening.base.forge_invention.inference	None
step:portfolio	contains	conceptual_frontier_deepening.base.portfolio.control	None
step:portfolio	contains	conceptual_frontier_deepening.base.portfolio.inference	None
step:reflect_discovery	contains	conceptual_frontier_deepening.base.reflect_discovery.control	None
step:reflect_discovery	contains	conceptual_frontier_deepening.base.reflect_discovery.inference	None
step:synthesize	contains	conceptual_frontier_deepening.base.synthesize.control	None
step:synthesize	contains	conceptual_frontier_deepening.base.synthesize.inference	None
step:synthesize	contains	conceptual_frontier_deepening.base.synthesize.inference	None

Source	Relation	Target	Adapter / detail
step:blind_adjudicate	contains	conceptual_frontier_deepening.base.blind_adjudicate.control	None
step:blind_adjudicate	contains	conceptual_frontier_deepening.base.blind_adjudicate.inference	None
step:compile_atlas_defects	contains	conceptual_frontier_deepening.base.compile_atlas_defects.control	None
step:compile_atlas_defects	contains	conceptual_frontier_deepening.base.compile_atlas_defects.inference	None
step:derive_central_routes	contains	conceptual_frontier_deepening.base.derive_central_routes.control	None
step:derive_central_routes	contains	conceptual_frontier_deepening.base.derive_central_routes.inference	None
step:check_bridge_equivalence	contains	conceptual_frontier_deepening.base.check_bridge_equivalence.control	None
step:check_bridge_equivalence	contains	conceptual_frontier_deepening.base.check_bridge_equivalence.inference	None
step:produce_formal_claim	contains	conceptual_frontier_deepening.base.produce_formal_claim.control	None
step:produce_formal_claim	contains	conceptual_frontier_deepening.base.produce_formal_claim.inference	None
step:plan_deepening	precedes	step:assimilate_literature	None
step:assimilate_literature	precedes	step:forge_invention	None
step:forge_invention	precedes	step:portfolio	None
step:portfolio	precedes	step:reflect_discovery	None
step:reflect_discovery	precedes	step:synthesize	None
step:synthesize	precedes	step:blind_adjudicate	None
step:blind_adjudicate	precedes	step:compile_atlas_defects	None
step:compile_atlas_defects	precedes	step:derive_central_routes	None
step:derive_central_routes	precedes	step:check_bridge_equivalence	None
step:check_bridge_equivalence	precedes	step:produce_formal_claim	None

H Projection completeness and omission ledger

Packet field	Disposition	Destination	Reason
alternatives_and_failures	rendered	negative-path body	Not recorded
artifacts_and_receipts	summarized_with_complete_appendix	artifact appendix	Not recorded
authority_ledger	rendered	authority body and appendix	Not recorded
claims	summarized_with_complete_appendix	claim ledger appendix	Not recorded
coverage	rendered	validation dashboard	Not recorded
decisions_and_gates	summarized_with_complete_appendix	decision and gate appendix	Not recorded
declared_route	rendered	workflow route body	Not recorded
evidence	summarized_with_complete_appendix	evidence ledger appendix	Not recorded

Packet field	Disposition	Destination	Reason
graph	summarized_with_complete_appendix	graph summary and complete edge appendix	Not recorded
handoffs	rendered	handoff body	Not recorded
identity	rendered	cover identity	Not recorded
inferences	rendered	expert inference dossiers	Not recorded
lens	rendered	cover and print profile	Not recorded
obligations	rendered	obligation body and appendix	Not recorded
profile_topology	rendered	route topology body	Not recorded
schema_version	rendered	build provenance	Not recorded
scope_ledger	rendered	scope body and appendix	Not recorded
specialized_projections	summarized_with_complete_appendix	specialized projection appendix	Not recorded
timeline	summarized_with_complete_appendix	complete chronology appendix	Not recorded

I Build provenance

Source packet lens

{"accent": "#c792ea", "focus_question": "Which conceptual hypothesis survived independent challenge and why?", "labels": {"claims": "Hypotheses and formal claims", "failures": "Rejected hypotheses and adjudication", "scope": "Frozen context and claim boundary"}, "lens_id": "conceptual_frontier_deepening", "mandatory_spine": ["freeze_context", "generate_hypotheses", "reflect_hypotheses", "blind_adjudicate", "formalize_claim", "audit_formal_claim"], "metrics": ["semantic_coverage", "evidence_resolution", "open_obligations", "trace_gaps"], "required_views": ["route", "inferences", "claims", "failures", "scope", "authority", "handoffs", "coverage"], "spine_label": "Frozen context to formal-ready claim", "visible_name": "Conceptual Frontier Deepening"}

Packet SHA-256

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Renderer version

1.0

Print inventory schema

1.0